Project 2

SEC Form 10-K & LLM-Sentiment-Based Portfolio Construction

The goal of this project is to test one or more strategies based on sentiment measures retrieved from SEC 10K filings and constructed via LLM.

Data Description

You can find the data in the Dropbox folders Sourcedata/SEC10X and Sourcedata/Project2:

- You will find lookup tables to link the different datasets together. These are stored in a folder named Sourcedata/Project2/lookup_tables.
- The folder Sourcedata/Project2/words_dictionary contains negative and positive opinion lexicon. Two files respectively contain a list of NEGATIVE and POSITIVE opinion words (or sentiment words). The files are retrieved from http://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html which is curated by the Computer Science Professor Bing Liu.
- You will find the Forms 10-K in the folder Sourcedata/SEC10X. Form 10-K files will be organized as follows: you will find folders for each year (e.g. 2018, 2019, 2020) with subfolders QTR1, QTR2, QTR3, QTR4 associated to each quarter. They will contain .txt files named as follows: 20180208_10-K_edgar_data_66740_0001558370-18-000535.txt The first 8 digits (20180208_10-K_...) denote the release date of the 10-K file while the bold digits …edgar_data_66740_... are associated to the CIK identifier of the 10-K filer (i.e. the company).
- To save you time, we created folders containing the 10-K files only for the years 2010 to 2021. You will need to extract 10-K files for 2022, 2023 and 2024.

Task Description

For each year *t*, please select the 500 firms associated with the largest market capitalizations as of December *t-1* (as you did in the momentum-SOLUTION.ipynb exercise; please use the same datasets as in the exercise).

Then, you will need to retrieve the 10-K forms associated with those firms.

You will extract the Item 7 section from each retrieved file. This section is called "Management's Discussion and Analysis of Financial Condition and Results of Operations". It gives the company's perspective on the business results of the past financial year. From this Item 7, construct sentiment indicators.

Then, based on these indicators, propose and test trading strategies.

Rather than relying on a dictionary-based approach, you will extract sentiment through open-source LLMs (Llama is suggested but please feel free to use any other alternatives).

Expected Deliverables

You should provide:

- a clear description of your sentiment-extraction methodology
- portfolio construction rules
- performance evaluation
- robustness checks
- economic interpretation of results
- what would the backtest if we observed the filings x days ahead of time? Interpret.

Tables and plots are strongly encouraged and programming codes must be submitted.